

Mockenhaupt's Three-Term Hardy–Littlewood Majorant Conjecture

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Abstract

For an integer $k \geq 0$, let $f_k(x) = 1 + e(x) + e((k+2)x)$ and $g_k(x) = 1 + e(x) - e((k+2)x)$ on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, where $e(x) = e^{2\pi i x}$. Mockenhaupt asked whether $\|g_k\|_{L^p(\mathbb{T})} > \|f_k\|_{L^p(\mathbb{T})}$ whenever $2k < p < 2k+2$; the later literature calls this Mockenhaupt's conjecture. The conjecture was previously known for $k \leq 5$. We give a single analytic proof valid for every $k \geq 4$; in particular, this settles all previously open cases $k \geq 6$ and establishes the conjecture for every $k \geq 0$.

The proof reduces the norm comparison to resonant Fourier coefficients on the two-torus and represents these coefficients, after analytic continuation, by triple-Bessel integrals. Neumann's product formula and the Weber–Schafheitlin formula yield a quantitative positive lower bound for the leading mode, while the remaining odd modes are controlled by a uniform tail estimate. The leading mode is then shown to dominate the tail for every $k \geq 4$.

AI Usage. The mathematical argument of this paper was produced by the author's personal AI research system. See Appendix A for the complete AI usage statement.

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1 Introduction

Write $e(x) = e^{2\pi i x}$ and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, equipped with normalized Lebesgue measure. For $0 < p < \infty$, we use

$$\|h\|_{L^p(\mathbb{T})} := \left(\int_{\mathbb{T}} |h(x)|^p dx \right)^{1/p},$$

with the usual quasi-norm interpretation when $p < 1$.

The Hardy–Littlewood majorant problem asks whether

$$\left\| \sum_{n \in \Lambda} a_n e(nx) \right\|_{L^p(\mathbb{T})} \leq \left\| \sum_{n \in \Lambda} b_n e(nx) \right\|_{L^p(\mathbb{T})}$$

whenever $\Lambda \subset \mathbb{Z}$ is finite and $|a_n| \leq b_n$ for every $n \in \Lambda$. For even integers $p = 2m$, the inequality follows from Parseval's identity applied to the m -th powers. For non-even exponents it fails in general; see [2, 3, 4, 8].

Montgomery conjectured that counterexamples also exist when the majorant is idempotent, meaning that its Fourier coefficients belong to $\{0, 1\}$, and the comparison polynomial is obtained by changing some of the nonzero coefficients to -1 . Mockenhaupt and Schlag established the

existence of such counterexamples using four-term idempotent polynomials [8]. Three is the smallest possible number of terms in such a counterexample: after multiplication by a character, a two-term idempotent has the form $1 + e(Mx)$, with $M \neq 0$, and

$$1 - e(Mx) = 1 + e\left(M\left(x + \frac{1}{2M}\right)\right).$$

Thus changing the relative sign preserves every L^p norm.

The pair $1 + e(x) \pm e(3x)$ already appears in the work of Hardy and Littlewood at $p = 3$; see also [4]. In his Habilitationsschrift [7], Mockenhaupt proposed a three-term polynomial of this kind for one range of exponents between consecutive even integers, and the later literature [4, 5, 6] formulated the same pattern for every such range. More precisely, for an integer $k \geq 0$, set

$$N := k + 2, \quad f_k(x) := 1 + e(x) + e(Nx), \quad g_k(x) := 1 + e(x) - e(Nx).$$

The statement known as Mockenhaupt's conjecture is that

$$\|g_k\|_{L^p(\mathbb{T})} > \|f_k\|_{L^p(\mathbb{T})} \quad (2k < p < 2k + 2). \quad (1)$$

The two coefficient sequences have the same absolute values and differ by a single sign.

The choice $N = k + 2$ is tied to the exponent interval. For an integer $1 \leq m < N$, every exponent in $(1 + z \pm z^N)^m$ has a unique representation $j + N\ell$, with $j, \ell \geq 0$ and $j + \ell \leq m$. Indeed, equality of two such representations implies

$$N|\ell - \ell'| = |j - j'| \leq m < N,$$

so $\ell = \ell'$ and $j = j'$. Changing the sign of z^N therefore preserves the absolute values of the coefficients of the m -th power. Parseval's identity gives

$$\|f_k\|_{L^{2m}(\mathbb{T})} = \|g_k\|_{L^{2m}(\mathbb{T})}, \quad 1 \leq m < N.$$

At $m = N$, the first collision occurs: the exponent N can arise either from a single factor z^N or from N factors z . For $k \geq 1$, the smallest third frequency for which both endpoints $2k$ and $2k + 2$ lie in the no-collision range is therefore $N = k + 2$. The conjecture is thus extremal in the number of terms and, in this sense, in the placement of the third frequency.

Krenedits proved the cases $k = 0, 1, 2$ in [4], the cases $k = 3, 4$ in [5], and the case $k = 5$ in [6]. The case $k = 0$ admits an analytic proof, while the cases $1 \leq k \leq 5$ rely on rigorous numerical estimates. We give a single analytic argument valid for every $k \geq 4$. In particular, this settles all previously open cases $k \geq 6$ and completes the proof of Mockenhaupt's conjecture.

A note on the name. What [7] contains is a problem, not a conjecture. The three-term inequality appears there in a footnote (pp. 32–33): for $m = 3$ and $2m - 2 < p < 2m$, the polynomial $z^m + z^{m+1} - 1$ is said to provide an example. Reflecting $n \mapsto -n$ and translating the frequencies by $m + 1$, neither of which changes any L^p norm, carries it to $1 + e(x) - e((m + 1)x)$, and the range becomes $(2k, 2k + 2)$ with $N = k + 2 = m + 1$. Thus the original asserts only the case $m = 3$, which is $k = 2$ here. The name *conjecture*, and the formulation for every k , come from the later literature [4, 5, 6]; we keep the name because that is how the statement is now known.

Theorem 1.1. *Let $k \geq 4$ be an integer. For every $2k < p < 2k + 2$,*

$$\|1 + e(x) - e((k + 2)x)\|_{L^p(\mathbb{T})} > \|1 + e(x) + e((k + 2)x)\|_{L^p(\mathbb{T})}.$$

Combining this with the known small cases gives the full conjecture.

Corollary 1.2. *Mockenhaupt's conjecture holds for every integer $k \geq 0$.*

Proof. The cases $k = 0, 1, 2$ follow from [4], and $k = 3$ follows from [5]. Theorem 1.1 covers all $k \geq 4$. $\square$

Proof strategy. Write $s = p/2 = k + \alpha$, with $0 < \alpha < 1$, and set

$$D(s) := \int_{\mathbb{T}} (|g_k(x)|^{2s} - |f_k(x)|^{2s}) dx.$$

We must prove $D(s) > 0$. Let $c_s(m, n)$ denote the Fourier coefficients of

$$H(\theta, \psi)^s := |1 + e(\theta) + e(\psi)|^{2s}.$$

Restriction to $(\theta, \psi) = (x, Nx)$ selects the modes $m + Nn = 0$. A shift by $1/2$ in the second variable changes the sign of precisely the odd modes, giving

$$D(s) = -4 \sum_{\substack{q \geq 1 \\ q \text{ odd}}} c_s(-Nq, q).$$

For $q \geq 1$, define

$$I_q(s) := \int_0^\infty J_{qN}(r) J_{q(N-1)}(r) J_q(r) r^{-2s-1} dr.$$

The Mellin representation of J_0 , followed by analytic continuation for each fixed mode, yields

$$D(s) = \frac{2^{2s+3} \Gamma(s+1)^2 \sin(\pi\alpha)}{\pi} \sum_{\substack{q \geq 1 \\ q \text{ odd}}} I_q(s). \quad (2)$$

The prefactor is positive. It remains to show that the leading mode dominates the sum of the remaining odd modes.

The leading mode I_1 is distinguished by the fact that its two large Bessel orders are adjacent. Neumann's product formula combines them with the nonnegative angular weight $\cos \vartheta$. The resulting integral is evaluated by the Weber–Schafheitlin formula, and Euler's transformation makes its positivity explicit. This gives a quantitative lower bound $I_1(s) \geq W_{\text{low}}(k, \alpha) > 0$. For the other odd modes, elementary Bessel bounds give

$$\sum_{\substack{q \geq 3 \\ q \text{ odd}}} |I_q(s)| \leq W_{\text{tail}}(k, \alpha).$$

The explicit bounds satisfy

$$\frac{W_{\text{tail}}(k, \alpha)}{W_{\text{low}}(k, \alpha)} < \frac{3}{4} \left(\frac{4}{5} \right)^{k-4} < 1, \quad k \geq 4, \quad 0 \leq \alpha \leq 1.$$

This proves $D(s) > 0$ throughout the required interval.

Throughout the paper, $X \asymp Y$ means that $cY \leq X \leq CY$ for some constants $c, C > 0$.

2 From the majorant problem to resonant Fourier modes

Fix

$$k \geq 4, \quad N := k + 2.$$

For real $s > 0$, set

$$A(x) := |f_k(x)|^2, \quad B(x) := |g_k(x)|^2,$$

and

$$D(s) := \int_{\mathbb{T}} (B(x)^s - A(x)^s) dx.$$

In the range relevant to Theorem 1.1, we write

$$s = \frac{p}{2} = k + \alpha, \quad 0 < \alpha < 1.$$

Then

$$D(s) = \|g_k\|_{L^p(\mathbb{T})}^p - \|f_k\|_{L^p(\mathbb{T})}^p. \quad (3)$$

Define

$$P(\theta, \psi) := 1 + e(\theta) + e(\psi), \quad H(\theta, \psi) := |P(\theta, \psi)|^2, \quad (\theta, \psi) \in \mathbb{T}^2.$$

Then

$$A(x) = H(x, Nx), \quad B(x) = H\left(x, Nx + \frac{1}{2}\right). \quad (4)$$

To justify restricting the Fourier series of H^s to the curves in (4) and integrating it term by term, we first record the local behavior of H at its zeros. Near a point of $\mathbb{T}^2$, distances are understood in local Euclidean coordinates.

Lemma 2.1. *The zeros of H are*

$$z_+ = \left(\frac{1}{3}, \frac{2}{3}\right), \quad z_- = \left(\frac{2}{3}, \frac{1}{3}\right).$$

Near either zero,

$$H(\theta, \psi) \asymp |(\theta, \psi) - z_{\pm}|^2. \quad (5)$$

Consequently, for every real σ ,

$$H^\sigma \in L^1(\mathbb{T}^2) \iff \sigma > -1. \quad (6)$$

Proof. The equation $P(\theta, \psi) = 0$ gives the two stated zeros. Regarding P as a map from $\mathbb{R}^2$ to $\mathbb{R}^2$, its Jacobian determinant at $z_{\pm}$ is $\pm 2\sqrt{3}\pi^2$, so P is a local diffeomorphism there. There exist $c, C, \delta > 0$ such that, for each $z \in \{z_+, z_-\}$,

$$c|(\theta, \psi) - z|^2 \leq H(\theta, \psi) \leq C|(\theta, \psi) - z|^2$$

whenever $|(\theta, \psi) - z| < \delta$. Thus (5) follows, and (6) follows from

$$\int_0^\delta r^{2\sigma+1} dr < \infty \iff \sigma > -1. \quad \square$$

The preceding local description gives the regularity needed for absolute convergence of the Fourier series of H^s . For $s \geq 1$ and $(m, n) \in \mathbb{Z}^2$, define

$$c_s(m, n) := \int_{\mathbb{T}^2} H(\theta, \psi)^s e(-m\theta - n\psi) d\theta d\psi.$$

Lemma 2.2. *For every real $s \geq 1$,*

$$\sum_{m, n \in \mathbb{Z}} |c_s(m, n)| < \infty.$$

Consequently,

$$H(\theta, \psi)^s = \sum_{m, n \in \mathbb{Z}} c_s(m, n) e(m\theta + n\psi) \quad (7)$$

absolutely and uniformly on $\mathbb{T}^2$.

Proof. Set

$$u(\theta, \psi) := H(\theta, \psi)^s.$$

Away from the two zeros of H , the function u is smooth. Near either zero, P provides smooth local coordinates in which

$$u = |w|^{2s}.$$

Since

$$D^2(|w|^{2s}) = O(|w|^{2s-2}),$$

the second derivatives of u are bounded near the zeros for $s \geq 1$. Therefore

$$u \in W^{2,2}(\mathbb{T}^2).$$

By the Fourier characterization of the Sobolev space $W^{2,2}(\mathbb{T}^2)$,

$$\sum_{m, n \in \mathbb{Z}} (1 + m^2 + n^2)^2 |c_s(m, n)|^2 < \infty.$$

Hence, by the Cauchy–Schwarz inequality,

$$\begin{aligned} \sum_{m, n \in \mathbb{Z}} |c_s(m, n)| &\leq \left(\sum_{m, n \in \mathbb{Z}} (1 + m^2 + n^2)^{-2} \right)^{1/2} \\ &\quad \times \left(\sum_{m, n \in \mathbb{Z}} (1 + m^2 + n^2)^2 |c_s(m, n)|^2 \right)^{1/2} < \infty. \end{aligned}$$

Hence the Fourier series converges absolutely and uniformly to a continuous function v . On the other hand, its partial sums converge to u in $L^2(\mathbb{T}^2)$. Thus $u = v$ almost everywhere. Since both u and v are continuous, they agree everywhere. This proves (7). $\square$

The absolute and uniform convergence above allows us to restrict the Fourier series of $H(\theta, \psi)^s$ to the two curves in (4) and integrate term by term.

Proposition 2.3 (Resonant Fourier modes). *For every real $s \geq 1$,*

$$D(s) = -4 \sum_{\substack{q \geq 1 \\ q \text{ odd}}} c_s(-Nq, q). \quad (8)$$

Proof. By Lemma 2.2,

$$H(\theta, \psi)^s = \sum_{m, n \in \mathbb{Z}} c_s(m, n) e(m\theta + n\psi)$$

absolutely and uniformly. Using (4),

$$\int_{\mathbb{T}} A(x)^s dx = \sum_{m, n \in \mathbb{Z}} c_s(m, n) \int_{\mathbb{T}} e((m + Nn)x) dx = \sum_{q \in \mathbb{Z}} c_s(-Nq, q).$$

Similarly,

$$\begin{aligned} \int_{\mathbb{T}} B(x)^s dx &= \sum_{m, n \in \mathbb{Z}} c_s(m, n) e(n/2) \int_{\mathbb{T}} e((m + Nn)x) dx \\ &= \sum_{q \in \mathbb{Z}} (-1)^q c_s(-Nq, q). \end{aligned}$$

Therefore

$$D(s) = -2 \sum_{\substack{q \in \mathbb{Z} \\ q \text{ odd}}} c_s(-Nq, q).$$

Since $H(-\theta, -\psi) = H(\theta, \psi)$, we have $c_s(-m, -n) = c_s(m, n)$, and (8) follows. $\square$

3 Bessel representation of the resonant modes

By Proposition 2.3, the norm difference is determined by the resonant Fourier coefficients $c_s(-Nq, q)$. In this section we represent these coefficients by triple-Bessel integrals. The identity is first established in a strip where the Mellin representation and Fubini's theorem apply, and is then continued holomorphically, for each fixed q , to the positive range required below.

Throughout this section, J_ν denotes the Bessel function of the first kind of order ν .

3.1 Holomorphic dependence on the exponent

For $\Re s > -1$, define

$$H(\theta, \psi)^s := \exp(s \log H(\theta, \psi))$$

when $H(\theta, \psi) > 0$, and set $H^s = 0$ at the zeros of H . For $(m, n) \in \mathbb{Z}^2$, let

$$c_s(m, n) := \int_{\mathbb{T}^2} H(\theta, \psi)^s e(-m\theta - n\psi) d\theta d\psi.$$

By Lemma 2.1, this integral is absolutely convergent. The convention at the zeros does not affect its value. For real $s \geq 1$, it agrees with the definition in Section 2.

Lemma 3.1. *For every $(m, n) \in \mathbb{Z}^2$, the function*

$$s \mapsto c_s(m, n)$$

is holomorphic in the half-plane

$$\Re s > -1.$$

Proof. Let $K \subseteq \{\Re s > -1\}$, and put

$$\sigma_0 := \min_{s \in K} \Re s, \quad \sigma_1 := \max_{s \in K} \Re s.$$

For $s \in K$ and $H > 0$,

$$|H^s| = H^{\Re s} \leq H^{\sigma_0} + H^{\sigma_1}.$$

The right-hand side is integrable by Lemma 2.1, since $\sigma_0 > -1$. Dominated convergence gives continuity of $c_s(m, n)$.

Let T be a triangle whose closure lies in this half-plane. The same bound, with $K = \partial T$, justifies Fubini's theorem:

$$\begin{aligned} \int_{\partial T} c_s(m, n) ds &= \int_{\mathbb{T}^2} e(-m\theta - n\psi) \left(\int_{\partial T} H(\theta, \psi)^s ds \right) d\theta d\psi \\ &= 0. \end{aligned}$$

The inner integral vanishes because $s \mapsto H^s$ is entire where $H > 0$, and is the constant zero function at the zeros of H under our convention. Morera's theorem proves the lemma. $\square$

3.2 The triple-Bessel coefficient

We next compute the Fourier coefficients of $J_0(r\sqrt{H})$ directly from the integral representation of J_0 .

Lemma 3.2. *For every $r > 0$ and $(m, n) \in \mathbb{Z}^2$,*

$$\int_{\mathbb{T}^2} J_0(r\sqrt{H(\theta, \psi)}) e(-m\theta - n\psi) d\theta d\psi = (-1)^{m+n} J_m(r) J_n(r) J_{m+n}(r). \quad (9)$$

In particular, for $q \geq 1$,

$$\int_{\mathbb{T}^2} J_0(r\sqrt{H}) e(Nq\theta - q\psi) d\theta d\psi = (-1)^{Nq} J_{qN}(r) J_{q(N-1)}(r) J_q(r). \quad (10)$$

Proof. Write $P(\theta, \psi) = 1 + e(\theta) + e(\psi)$, so that $\sqrt{H} = |P|$. The integral representation of J_0 gives

$$J_0(r|P|) = \frac{1}{2\pi} \int_0^{2\pi} \exp(ir\Re(e^{-it}P)) dt.$$

Since

$$\Re(e^{-it}P) = \cos t + \cos(2\pi\theta - t) + \cos(2\pi\psi - t),$$

Fubini's theorem gives

$$\int_{\mathbb{T}^2} J_0(r\sqrt{H})e(-m\theta - n\psi)d\theta d\psi = \frac{1}{2\pi} \int_0^{2\pi} e^{ir \cos t} A_m(t) A_n(t) dt,$$

where

$$A_m(t) := \int_{\mathbb{T}} e^{ir \cos(2\pi\theta - t)} e(-m\theta) d\theta.$$

The interchange is justified because the integrand has modulus 1 on the finite-measure product space.

For each integer ℓ , Bessel's integral formula [1, Eq. 10.9.2] gives

$$\frac{1}{2\pi} \int_0^{2\pi} e^{ir \cos u} e^{-i\ell u} du = i^\ell J_\ell(r).$$

The change of variables $u = 2\pi\theta - t$, followed by periodicity, therefore gives $A_m(t) = i^m J_m(r) e^{-imt}$. Consequently,

$$\begin{aligned} \int_{\mathbb{T}^2} J_0(r\sqrt{H})e(-m\theta - n\psi)d\theta d\psi &= i^{m+n} J_m(r) J_n(r) \frac{1}{2\pi} \int_0^{2\pi} e^{ir \cos t} e^{-i(m+n)t} dt \\ &= (-1)^{m+n} J_m(r) J_n(r) J_{m+n}(r). \end{aligned}$$

Taking $(m, n) = (-Nq, q)$ and using $J_{-\ell} = (-1)^\ell J_\ell$ gives (10). □

3.3 Mode-by-mode analytic continuation

For $q \geq 1$, set

$$\Psi_q(r) := J_{qN}(r) J_{q(N-1)}(r) J_q(r),$$

and define

$$I_q(s) := \int_0^\infty \Psi_q(r) r^{-2s-1} dr \tag{11}$$

whenever the integral is absolutely convergent. We first establish a strip of absolute convergence in which the mode-by-mode continuation will be carried out.

Lemma 3.3. *For $q \geq 1$, set*

$$C_q := \Gamma(qN + 1) \Gamma(q(N - 1) + 1) \Gamma(q + 1).$$

Then

$$|\Psi_q(r)| \leq \frac{(r/2)^{2qN}}{C_q}, \quad r > 0, \tag{12}$$

and, for each fixed q ,

$$\Psi_q(r) = O_q(r^{-3/2}) \quad (r \rightarrow \infty).$$

Consequently, I_q is holomorphic on

$$\Omega_q := \left\{ s \in \mathbb{C} : -\frac{3}{4} < \Re s < qN \right\}.$$

Proof. The three orders satisfy

$$qN + q(N - 1) + q = 2qN.$$

Poisson's integral representation [1, Eq. 10.9.4] gives, for $\ell \geq 0$,

$$J_\ell(r) = \frac{2(r/2)^\ell}{\sqrt{\pi} \Gamma(\ell + \frac{1}{2})} \int_0^1 (1 - t^2)^{\ell - \frac{1}{2}} \cos(rt) dt.$$

Hence

$$|J_\ell(r)| \leq \frac{2(r/2)^\ell}{\sqrt{\pi} \Gamma(\ell + \frac{1}{2})} \int_0^1 (1 - t^2)^{\ell - \frac{1}{2}} dt.$$

Since

$$\int_0^1 (1 - t^2)^{\ell - \frac{1}{2}} dt = \frac{\sqrt{\pi} \Gamma(\ell + \frac{1}{2})}{2\Gamma(\ell + 1)},$$

we obtain

$$|J_\ell(r)| \leq \frac{(r/2)^\ell}{\Gamma(\ell + 1)}, \quad r > 0.$$

Multiplying the three estimates gives (12). For each fixed ℓ , the large-argument estimate [1, Eq. 10.17.3]

$$J_\ell(r) = O_\ell(r^{-1/2}) \quad (r \rightarrow \infty)$$

gives $\Psi_q(r) = O_q(r^{-3/2})$.

Let $K \Subset \Omega_q$, and set

$$\sigma_0 := \min_{s \in K} \Re s, \quad \sigma_1 := \max_{s \in K} \Re s.$$

Then $\sigma_0 > -3/4$ and $\sigma_1 < qN$. For $s \in K$, (12) gives

$$\left| \Psi_q(r) r^{-2s-1} \right| \leq \frac{2^{-2qN}}{C_q} r^{2qN - 2\sigma_1 - 1}, \quad 0 < r \leq 1,$$

whereas the large-argument bound gives

$$\left| \Psi_q(r) r^{-2s-1} \right| \leq C_{q,K} r^{-2\sigma_0 - 5/2}, \quad r \geq 1.$$

Both majorants are integrable. Hence $I_q(s)$ is absolutely convergent on Ω_q , locally uniformly in s . Since $s \mapsto r^{-2s-1}$ is entire for each $r > 0$, the standard holomorphic parameter-integral theorem shows that I_q is holomorphic on Ω_q . $\square$

The Mellin integral representing $H(\theta, \psi)^s$ is absolutely convergent only in the strip

$$-\frac{1}{4} < \Re s < 0.$$

After taking the q -th resonant Fourier coefficient, however, the triple-Bessel product vanishes to order $2qN$ at the origin. Consequently, $I_q(s)$ is holomorphic on the larger strip Ω_q , which contains the positive exponents required below. This allows the identity to be continued mode by mode.

Set

$$C(s) := \frac{2^{2s+1}\Gamma(s+1)}{\Gamma(-s)}.$$

The precise relation between $c_s(-Nq, q)$ and $I_q(s)$ is the following.

Proposition 3.4. *For every $q \geq 1$ and every $s \in \Omega_q$,*

$$c_s(-Nq, q) = C(s)(-1)^{Nq}I_q(s). \quad (13)$$

Proof. We begin in the strip

$$-\frac{1}{4} < \Re s < 0.$$

There the Mellin formula for J_0 [1, Eq. 10.22.43] gives

$$a^{2s} = C(s) \int_0^\infty J_0(ar) r^{-2s-1} dr, \quad a > 0. \quad (14)$$

Taking $a = \sqrt{H(\theta, \psi)}$, we obtain

$$H(\theta, \psi)^s = C(s) \int_0^\infty J_0(r\sqrt{H(\theta, \psi)}) r^{-2s-1} dr$$

whenever $H(\theta, \psi) > 0$, and hence almost everywhere on $\mathbb{T}^2$.

It remains to justify the change in the order of integration. Put $\sigma = \Re s$ and

$$M(\sigma) := \int_0^\infty |J_0(t)| t^{-2\sigma-1} dt.$$

The estimates $J_0(t) = O(1)$ as $t \rightarrow 0$ and $J_0(t) = O(t^{-1/2})$ as $t \rightarrow \infty$ show that $M(\sigma) < \infty$ when $-1/4 < \sigma < 0$. Let

$$Z := \{(\theta, \psi) \in \mathbb{T}^2 : H(\theta, \psi) = 0\}.$$

For $(\theta, \psi) \notin Z$, the change of variables $t = r\sqrt{H}$ gives

$$\int_0^\infty |J_0(r\sqrt{H})| r^{-2\sigma-1} dr = H^\sigma M(\sigma).$$

Consequently, Lemma 2.1 gives the absolute bound

$$\int_{\mathbb{T}^2 \setminus Z} \int_0^\infty |J_0(r\sqrt{H})| r^{-2\sigma-1} dr d\theta d\psi = M(\sigma) \int_{\mathbb{T}^2} H^\sigma d\theta d\psi < \infty.$$

Thus Fubini's theorem applies. Since Z has measure zero, it may be restored in the inner integral after the interchange, and we obtain

$$\begin{aligned} c_s(-Nq, q) &= C(s) \int_0^\infty r^{-2s-1} \left(\int_{\mathbb{T}^2} J_0(r\sqrt{H}) e(Nq\theta - q\psi) d\theta d\psi \right) dr \\ &= C(s)(-1)^{Nq} I_q(s), \end{aligned}$$

where the last equality follows from Lemma 3.2. Hence (13) holds in the anchor strip $-1/4 < \Re s < 0$.

By Lemma 3.1, $c_s(-Nq, q)$ is holomorphic on Ω_q , while Lemma 3.3 gives the same for $I_q(s)$. Moreover,

$$C(s) = \frac{2^{2s+1}\Gamma(s+1)}{\Gamma(-s)}$$

is holomorphic there: $1/\Gamma(-s)$ is entire, and $\Gamma(s+1)$ has no pole in $\Re s > -3/4$. Since Ω_q is connected and contains the anchor strip, the identity theorem extends (13) to all of Ω_q . $\square$

Theorem 3.5 (Bessel representation). *Let*

$$k \geq 4, \quad s = k + \alpha, \quad 0 < \alpha < 1.$$

Then

$$D(s) = \frac{2^{2s+3}\Gamma(s+1)^2 \sin(\pi\alpha)}{\pi} \sum_{\substack{q \geq 1 \\ q \text{ odd}}} I_q(s), \quad (15)$$

and the series converges absolutely.

Proof. Since

$$s < k + 1 = N - 1 < N \leq qN,$$

we have $s \in \Omega_q$ for every $q \geq 1$, so Proposition 3.4 applies to every resonant mode. Moreover, if q is odd, then

$$(-1)^{Nq} = (-1)^N = (-1)^k.$$

Proposition 2.3 therefore gives

$$D(s) = -4C(s)(-1)^k \sum_{\substack{q \geq 1 \\ q \text{ odd}}} I_q(s).$$

By the reflection formula,

$$\frac{1}{\Gamma(-s)} = -\frac{\Gamma(s+1) \sin(\pi s)}{\pi},$$

and hence

$$C(s) = -\frac{2^{2s+1}\Gamma(s+1)^2 \sin(\pi s)}{\pi}.$$

Since $\sin(\pi s) = \sin(\pi(k + \alpha)) = (-1)^k \sin(\pi\alpha)$, we obtain (15).

Finally, $0 < \alpha < 1$ implies $C(s) \neq 0$. Using (13) and the absolute summability of the Fourier coefficients from Lemma 2.2, we have

$$\sum_{\substack{q \geq 1 \\ q \text{ odd}}} |I_q(s)| = \frac{1}{|C(s)|} \sum_{\substack{q \geq 1 \\ q \text{ odd}}} |c_s(-Nq, q)| \leq \frac{1}{|C(s)|} \sum_{m, n \in \mathbb{Z}} |c_s(m, n)| < \infty. \quad \square$$

The prefactor in (15) is positive. Thus the proof of Theorem 1.1 is reduced to showing that

$$\sum_{\substack{q \geq 1 \\ q \text{ odd}}} I_q(s) > 0. \quad (16)$$

4 The leading mode

Throughout this section,

$$k \geq 4, \quad N = k + 2, \quad 0 \leq \alpha \leq 1, \quad s = k + \alpha.$$

We consider the leading term

$$I_1(s) = \int_0^\infty J_N(r) J_{N-1}(r) J_1(r) r^{-2s-1} dr.$$

Set

$$L := 2N - 1 = 2k + 3.$$

The adjacent orders N and $N - 1$ allow us to use Neumann's product formula with the nonnegative weight $\cos \vartheta$. This reduces the positivity question to an integral of two Bessel functions, which we evaluate below.

Proposition 4.1. *For $0 \leq \alpha \leq 1$,*

$$I_1(s) = \frac{2}{\pi} \int_0^{\pi/2} \cos \vartheta W(2 \cos \vartheta) d\vartheta, \quad (17)$$

where

$$W(A) := \int_0^\infty J_L(Ar) J_1(r) r^{-2s-1} dr, \quad 0 \leq A \leq 2.$$

Proof. Neumann's product formula [1, Eq. 10.9.26] (see also [9, Eq. (5.43.1), p. 150]) gives

$$J_N(r) J_{N-1}(r) = \frac{2}{\pi} \int_0^{\pi/2} J_L(2r \cos \vartheta) \cos \vartheta d\vartheta.$$

Its order condition is automatic here, since $N + (N - 1) > -1$.

We verify absolute integrability before changing the order of integration. For $0 < r \leq 1$, uniformly in $0 \leq \vartheta \leq \pi/2$,

$$|J_L(2r \cos \vartheta)| \leq \frac{r^L}{\Gamma(L + 1)}, \quad |J_1(r)| \leq \frac{r}{2}.$$

Hence

$$\left| J_L(2r \cos \vartheta) J_1(r) \cos \vartheta r^{-2s-1} \right| \leq \frac{1}{2\Gamma(L + 1)} r^{L-2s} = \frac{1}{2\Gamma(L + 1)} r^{3-2\alpha},$$

which is integrable at 0 for $0 \leq \alpha \leq 1$. For $r \geq 1$, the bound [1, Eq. 10.14.1]

$$|J_n(x)| \leq 1, \quad x \in \mathbb{R}, \quad n = 0, 1, 2, \dots,$$

gives

$$\left| J_L(2r \cos \vartheta) J_1(r) \cos \vartheta r^{-2s-1} \right| \leq r^{-2s-1},$$

which is integrable at infinity. Thus the corresponding integral over $(0, \infty) \times (0, \pi/2)$ is absolutely convergent, and Fubini's theorem yields (17). $\square$

To evaluate $W(A)$, we use the following form of the Weber–Schafheitlin formula [1, Eq. 10.22.56] (see also [9, Section 13.4]). If

$$0 < a < b, \quad \Re(\mu + \nu + 1) > \Re \lambda > -1,$$

then

$$\begin{aligned} \int_0^\infty J_\mu(ar) J_\nu(br) r^{-\lambda} dr &= \frac{a^\mu \Gamma\left(\frac{\mu+\nu-\lambda+1}{2}\right)}{2^\lambda b^{\mu-\lambda+1} \Gamma\left(\frac{\nu-\mu+\lambda+1}{2}\right) \Gamma(\mu+1)} \\ &\quad \times {}_2F_1\left(\frac{\mu+\nu-\lambda+1}{2}, \frac{\mu-\nu-\lambda+1}{2}; \mu+1; \frac{a^2}{b^2}\right). \end{aligned} \quad (18)$$

Here

$${}_2F_1(a, b; c; z) = \sum_{j=0}^{\infty} \frac{(a)_j (b)_j}{(c)_j j!} z^j, \quad |z| < 1,$$

where

$$(a)_0 = 1, \quad (a)_j = a(a+1) \cdots (a+j-1), \quad j \geq 1.$$

Formula (18) is [1, Eq. 10.22.56] written in terms of the usual Gauss hypergeometric function. Indeed, the DLMF uses Olver's normalized function

$$\mathbf{F}(a, b; c; z) = \frac{{}_2F_1(a, b; c; z)}{\Gamma(c)}$$

(cf. [1, Eq. 15.1.2]), which accounts for the factor $\Gamma(\mu+1)^{-1}$ in (18).

Proposition 4.2. *Let $0 \leq \alpha \leq 1$ and $s = k + \alpha$. If $0 < A < 1$, then*

$$W(A) = \frac{A^L \Gamma(2-\alpha)}{2^{2s+1} \Gamma(\alpha) \Gamma(L+1)} {}_2F_1\left(2-\alpha, 1-\alpha; L+1; A^2\right), \quad (19)$$

where the value at $\alpha = 0$ is understood by continuity.

If $1 < A \leq 2$, then

$$W(A) = \frac{\Gamma(2-\alpha) A^{2s-1}}{2^{2s+1} \Gamma(2k+2+\alpha)} (1-A^{-2})^{2s+1} {}_2F_1\left(\alpha, 2k+3+\alpha; 2; A^{-2}\right). \quad (20)$$

Proof. In both applications of (18), the orders are $L = 2k+3$ and 1, and $\lambda = 2s+1$. Thus

$$\mu + \nu + 1 - \lambda = 4 - 2\alpha > 0, \quad \lambda > -1,$$

so the order conditions hold for $0 \leq \alpha \leq 1$.

For $0 < A < 1$, take $(a, b, \mu, \nu, \lambda) = (A, 1, L, 1, 2s + 1)$. The two upper hypergeometric parameters are $2 - \alpha$ and $1 - \alpha$; the lower parameter is $L + 1$, and the other Gamma factor in the denominator is $\Gamma(\alpha)$. This gives (19) for $0 < \alpha \leq 1$. At $\alpha = 0$, the defining integral for $W(A)$ is continuous in α : its integrand is bounded uniformly by Cr on $0 < r \leq 1$ and by r^{-2k-1} on $r \geq 1$. For fixed $A < 1$, the hypergeometric factor remains finite as $\alpha \rightarrow 0^+$, while $1/\Gamma(\alpha) \rightarrow 0$. Hence $W(A) = 0$ at $\alpha = 0$.

For $1 < A \leq 2$, interchange the two Bessel factors and take $(a, b, \mu, \nu, \lambda) = (1, A, 1, L, 2s + 1)$. Formula (18) gives

$$W(A) = \frac{\Gamma(2 - \alpha)A^{2s-1}}{2^{2s+1}\Gamma(2k + 2 + \alpha)} {}_2F_1\left(2 - \alpha, -2k - 1 - \alpha; 2; A^{-2}\right).$$

Apply Euler's transformation [1, Eq. 15.8.1]

$${}_2F_1(a, b; c; x) = (1 - x)^{c-a-b} {}_2F_1(c - a, c - b; c; x), \quad 0 < x < 1,$$

with $a = 2 - \alpha$, $b = -2k - 1 - \alpha$ and $c = 2$. Since

$$c - a = \alpha, \quad c - b = 2k + 3 + \alpha, \quad c - a - b = 2s + 1,$$

we obtain (20). □

The value $A = 1$ does not affect (17), since $A = 1$ corresponds to the single point $\vartheta = \pi/3$. We use (19) to show that the contribution from $0 < A < 1$ is nonnegative, and (20) to obtain a quantitative lower bound from the region $1 < A < 2$.

Proposition 4.3. *For every $0 \leq \alpha \leq 1$,*

$$I_1(s) \geq W_{\text{low}}(k, \alpha) > 0, \tag{21}$$

where

$$W_{\text{low}}(k, \alpha) := \frac{\Gamma(2 - \alpha)}{\pi\Gamma(2k + 2 + \alpha)(2s + 2)} \cdot \frac{2}{5\sqrt{3}} \cdot \frac{9}{8} \left(\frac{9}{16}\right)^s. \tag{22}$$

Proof. For $0 < \alpha \leq 1$, all factors in (19) are nonnegative when $0 < A < 1$: the two upper parameters of the hypergeometric series are $2 - \alpha > 0$ and $1 - \alpha \geq 0$, while its lower parameter is $L + 1 > 0$. At $\alpha = 0$, the limiting value is $W(A) = 0$. Thus

$$W(A) \geq 0, \quad 0 < A < 1,$$

and the corresponding part of (17) may be discarded.

For $1 < A < 2$, every coefficient of the hypergeometric series in (20) is nonnegative. Hence

$${}_2F_1\left(\alpha, 2k + 3 + \alpha; 2; A^{-2}\right) \geq 1.$$

Using $A = 2 \cos \vartheta$ in (17) and retaining only the region $1 < A < 2$, we obtain

$$I_1(s) \geq \frac{\Gamma(2 - \alpha)}{\pi 2^{2s+1}\Gamma(2k + 2 + \alpha)} \int_1^2 \frac{A^{2s}(1 - A^{-2})^{2s+1}}{\sqrt{4 - A^2}} dA.$$

Now set

$$t := A - \frac{1}{A}.$$

Since

$$A^{2s}(1 - A^{-2})^{2s+1} = \frac{t^{2s+1}}{A}, \quad \frac{dA}{A} = \frac{A}{A^2 + 1} dt,$$

we have

$$\int_1^2 \frac{A^{2s}(1 - A^{-2})^{2s+1}}{\sqrt{4 - A^2}} dA = \int_0^{3/2} t^{2s+1} \frac{A}{A^2 + 1} \frac{dt}{\sqrt{4 - A^2}}.$$

For $1 < A < 2$,

$$\frac{A}{A^2 + 1} \geq \frac{2}{5}, \quad \frac{1}{\sqrt{4 - A^2}} \geq \frac{1}{\sqrt{3}},$$

and therefore

$$\int_1^2 \frac{A^{2s}(1 - A^{-2})^{2s+1}}{\sqrt{4 - A^2}} dA \geq \frac{2}{5\sqrt{3}} \cdot \frac{(3/2)^{2s+2}}{2s + 2}.$$

Finally,

$$\frac{(3/2)^{2s+2}}{2^{2s+1}} = \frac{9}{8} \left(\frac{9}{16} \right)^s,$$

which gives (21). □

We have obtained a positive quantitative lower bound for the leading mode. The next section estimates the sum of the remaining odd modes.

5 The tail

Section 4 gives a positive lower bound for the leading mode. We now estimate the remaining odd modes:

$$\sum_{\substack{q \geq 3 \\ q \text{ odd}}} |I_q(s)|.$$

Throughout this section,

$$k \geq 4, \quad N = k + 2 \geq 6, \quad s = k + \alpha, \quad 0 \leq \alpha \leq 1.$$

In particular,

$$0 < s \leq N - 1 < qN, \quad q \geq 1.$$

Recall that

$$C_q = \Gamma(qN + 1)\Gamma(q(N - 1) + 1)\Gamma(q + 1),$$

and set

$$R_q := 2C_q^{1/(2qN)}.$$

By (12),

$$|\Psi_q(r)| \leq \left(\frac{r}{R_q}\right)^{2qN}.$$

On the other hand, the standard bound

$$|J_\ell(r)| \leq 1, \quad \ell = 0, 1, 2, \dots,$$

gives $|\Psi_q(r)| \leq 1$. Hence

$$|\Psi_q(r)| \leq \min \left\{ 1, \left(\frac{r}{R_q}\right)^{2qN} \right\}.$$

Lemma 5.1. *For every integer $q \geq 1$,*

$$|I_q(s)| \leq R_q^{-2s} \left(\frac{1}{2qN - 2s} + \frac{1}{2s} \right).$$

Proof. Since $s > 0$ and $qN > s$, splitting the integral at R_q gives

$$\begin{aligned} |I_q(s)| &\leq \int_0^{R_q} \left(\frac{r}{R_q}\right)^{2qN} r^{-2s-1} dr + \int_{R_q}^\infty r^{-2s-1} dr \\ &= R_q^{-2s} \left(\frac{1}{2qN - 2s} + \frac{1}{2s} \right). \end{aligned}$$

□

To make the q -dependence explicit, we bound R_q linearly from below.

Lemma 5.2. *For every $N \geq 6$ and every integer $q \geq 1$,*

$$R_q \geq \frac{29}{50} Nq.$$

Proof. Using $n! \geq \left(\frac{n}{e}\right)^n$, ($n \geq 1$), we obtain

$$C_q^{1/(2qN)} \geq \frac{q}{e} N^{1/2} (N-1)^{(N-1)/(2N)}.$$

Hence

$$\frac{R_q}{Nq} \geq \frac{2}{e} \left(\frac{(1 - 1/N)^{N-1}}{N} \right)^{1/(2N)} \geq \frac{2}{e} (eN)^{-1/(2N)},$$

where we use

$$\left(1 - \frac{1}{N}\right)^{N-1} \geq e^{-1}.$$

Set

$$F(N) := \frac{2}{e} (eN)^{-1/(2N)}.$$

Since

$$\frac{d}{dN} \log F(N) = \frac{\log N}{2N^2} > 0,$$

F is increasing for $N > 1$. Thus it is enough to check $N = 6$.

Since

$$F(6) > \frac{29}{50} \iff 6 \left(\frac{29}{50} \right)^{12} e^{13} < 2^{12},$$

it remains to verify the latter inequality. First,

$$e = \sum_{j=0}^{\infty} \frac{1}{j!} < \frac{65}{24} + \frac{1}{120} \sum_{j=0}^{\infty} 6^{-j} = \frac{65}{24} + \frac{1}{100} < \frac{68}{25}.$$

Moreover,

$$\left(\frac{29}{50} \right)^{12} < \frac{3}{2000}, \quad \left(\frac{68}{25} \right)^{13} < 450000,$$

which follow, for instance, from

$$\left(\frac{29}{50} \right)^4 < \frac{57}{500}, \quad \left(\frac{68}{25} \right)^4 < \frac{219}{4}.$$

Consequently,

$$6 \left(\frac{29}{50} \right)^{12} e^{13} < 6 \cdot \frac{3}{2000} \cdot 450000 = 4050 < 4096 = 2^{12}.$$

Hence $F(6) > \frac{29}{50}$. This proves the lemma. □

Set

$$\kappa := \frac{29}{50}.$$

For the sum over odd integers $q \geq 3$, set

$$B_N(s) := \frac{1}{6N - 2s} + \frac{1}{2s}, \tag{23}$$

and

$$T(s) := 1 + \frac{3}{2(2s - 1)}. \tag{24}$$

Define

$$W_{\text{tail}}(k, \alpha) := (\kappa N)^{-2s} B_N(s) 3^{-2s} T(s). \tag{25}$$

Proposition 5.3. *For $k \geq 4$ and $0 \leq \alpha \leq 1$,*

$$\sum_{\substack{q \geq 3 \\ q \text{ odd}}} |I_q(s)| \leq W_{\text{tail}}(k, \alpha). \tag{26}$$

Proof. For $q \geq 3$,

$$2qN - 2s \geq 6N - 2s > 0.$$

Hence Lemmas 5.1 and 5.2 give

$$|I_q(s)| \leq (\kappa N)^{-2s} B_N(s) q^{-2s}.$$

Since x^{-2s} is decreasing,

$$\begin{aligned} \sum_{\substack{q \geq 3 \\ q \text{ odd}}} q^{-2s} &\leq 3^{-2s} + \frac{1}{2} \int_3^\infty x^{-2s} dx \\ &= 3^{-2s} \left(1 + \frac{3}{2(2s-1)} \right) = 3^{-2s} T(s). \end{aligned}$$

Combining the two estimates proves (26). □

By (21) and (26), it remains to prove

$$W_{\text{tail}}(k, \alpha) < W_{\text{low}}(k, \alpha), \quad k \geq 4, \quad 0 \leq \alpha \leq 1. \quad (27)$$

This is proved in Section 6.

6 The leading mode dominates the tail

We compare the lower bound from Section 4 with the upper bound from Section 5. Recall that

$$\kappa = \frac{29}{50}, \quad N = k + 2, \quad s = k + \alpha, \quad 0 \leq \alpha \leq 1.$$

Set

$$R(k, \alpha) := \frac{W_{\text{tail}}(k, \alpha)}{W_{\text{low}}(k, \alpha)}.$$

Then (27) is equivalent to

$$R(k, \alpha) < 1.$$

By the definitions of W_{low} and W_{tail} ,

$$R(k, \alpha) = \frac{20\sqrt{3}\pi}{9} \frac{\Gamma(2k+2+\alpha)}{\Gamma(2-\alpha)} (2s+2) B_N(s) T(s) \left(\frac{16}{81\kappa^2 N^2} \right)^s.$$

Since

$$2k+2+\alpha = s+N, \quad 2-\alpha = N-s,$$

this becomes

$$R(k, \alpha) = \frac{20\sqrt{3}\pi}{9} \frac{\Gamma(s+N)}{\Gamma(N-s)} (2s+2) B_N(s) T(s) \left(\frac{16}{81\kappa^2 N^2} \right)^s.$$

Set

$$C_0 := \frac{20\sqrt{3}\pi}{9}, \quad \mu_N := \frac{16}{81\kappa^2 N^2}.$$

Thus

$$R(k, \alpha) = C_0 \frac{\Gamma(s+N)}{\Gamma(N-s)} (2s+2) B_N(s) T(s) \mu_N^s, \quad N-2 \leq s \leq N-1. \quad (28)$$

We first show that $R(k, \alpha)$ is decreasing in α .

Lemma 6.1. *For every integer $k \geq 4$,*

$$R(k, \alpha) \leq R(k, 0), \quad 0 \leq \alpha \leq 1.$$

Proof. Since $s = k + \alpha$, differentiation in α is the same as differentiation in s , with N fixed. By (28),

$$\begin{aligned} \frac{d}{ds} \log R(k, \alpha) &= \psi(s+N) + \psi(N-s) + \log \mu_N + \frac{1}{s+1} \\ &\quad + \frac{B'_N(s)}{B_N(s)} + \frac{T'(s)}{T(s)}, \end{aligned}$$

where $\psi = \Gamma'/\Gamma$. On $N-2 \leq s \leq N-1$,

$$B'_N(s) = \frac{2}{(6N-2s)^2} - \frac{2}{(2s)^2} < 0, \quad T'(s) = -\frac{3}{(2s-1)^2} < 0.$$

Strict convexity of $\log \Gamma$ gives, for $x > 0$,

$$\psi(x) < \log \Gamma(x+1) - \log \Gamma(x) = \log x,$$

where we used $\Gamma(x+1) = x\Gamma(x)$. Using also $(s+N)(N-s) = N^2 - s^2 \leq 4N - 4 < 4N$, we obtain

$$\begin{aligned} \frac{d}{ds} \log R(k, \alpha) &< \log(\mu_N(s+N)(N-s)) + \frac{1}{s+1} \\ &< \log\left(\frac{64}{81\kappa^2 N}\right) + \frac{1}{N-1} \\ &\leq \log \frac{2}{5} + \frac{1}{5} < 0. \end{aligned}$$

Indeed, $N \geq 6$ implies

$$\frac{64}{81\kappa^2 N} \leq \frac{80000}{204363} < \frac{2}{5}, \quad \frac{1}{N-1} \leq \frac{1}{5},$$

and the last inequality follows from $\log x \leq x - 1$. □

It remains to estimate the endpoint value $R(k, 0)$. Set

$$B_0(k) := \frac{1}{4k+12} + \frac{1}{2k}, \quad T_0(k) := 1 + \frac{3}{2(2k-1)}.$$

At $\alpha = 0$,

$$s = k, \quad N - s = 2, \quad s + N = 2k + 2,$$

and hence

$$\Gamma(N - s) = 1, \quad (2s + 2)\Gamma(s + N) = \Gamma(2k + 3).$$

Thus (28) becomes

$$R(k, 0) = C_0 \Gamma(2k + 3) B_0(k) T_0(k) \left(\frac{16}{81\kappa^2(k + 2)^2} \right)^k. \quad (29)$$

Lemma 6.2. *For every integer $k \geq 4$,*

$$R(k, 0) < \frac{3}{4} \left(\frac{4}{5} \right)^{k-4}.$$

Proof. For $k = 4$,

$$\Gamma(11) = 10!, \quad B_0(4) = \frac{9}{56} < \frac{1}{6}, \quad T_0(4) = \frac{17}{14} < \frac{5}{4}.$$

Also, using $\pi < 22/7$ and $\sqrt{3} < 7/4$,

$$C_0 < \frac{25}{2}, \quad \mu_6 = \frac{10000}{613089} < \frac{1}{61}.$$

Hence

$$R(4, 0) < \frac{25}{2} \cdot 10! \cdot \frac{1}{6} \cdot \frac{5}{4} \cdot \frac{1}{61^4} = \frac{9450000}{13845841} < \frac{3}{4}.$$

For $k \geq 4$, formula (29) gives

$$\frac{R(k + 1, 0)}{R(k, 0)} = \frac{(2k + 4)(2k + 3)}{(k + 3)^2} \frac{16}{81\kappa^2} \left(1 - \frac{1}{k + 3} \right)^{2k} \frac{B_0(k + 1)}{B_0(k)} \frac{T_0(k + 1)}{T_0(k)}.$$

Since B_0 and T_0 are positive and decreasing,

$$\frac{(2k + 4)(2k + 3)}{(k + 3)^2} < 4, \quad \frac{16}{81\kappa^2} = \frac{40000}{68121} < \frac{3}{5},$$

and

$$\left(1 - \frac{1}{k + 3} \right)^{2k} \leq e^{-2k/(k+3)} \leq e^{-8/7} < \frac{1}{3}.$$

The last inequality follows, for instance, from

$$e^{8/7} > 1 + \frac{8}{7} + \frac{1}{2} \left(\frac{8}{7} \right)^2 + \frac{1}{6} \left(\frac{8}{7} \right)^3 > 3.$$

Therefore

$$\frac{R(k + 1, 0)}{R(k, 0)} < 4 \cdot \frac{3}{5} \cdot \frac{1}{3} = \frac{4}{5}.$$

Combining this with the estimate at $k = 4$ proves the lemma. $\square$

Combining the two lemmas, we obtain

$$R(k, \alpha) \leq R(k, 0) < \frac{3}{4} \left(\frac{4}{5} \right)^{k-4} < 1.$$

Thus

$$W_{\text{tail}}(k, \alpha) < W_{\text{low}}(k, \alpha).$$

Together with (21) and (26), this gives

$$\sum_{\substack{q \geq 3 \\ q \text{ odd}}} |I_q(s)| < I_1(s).$$

The same estimates give the quantitative approximation

$$\sum_{\substack{q \geq 1 \\ q \text{ odd}}} I_q(s) = I_1(s)(1 + \eta_{k, \alpha}), \quad |\eta_{k, \alpha}| < \frac{3}{4} \left(\frac{4}{5} \right)^{k-4}. \quad (30)$$

The relative error bound is uniform for $0 \leq \alpha \leq 1$.

Proof of Theorem 1.1. Let $k \geq 4$, $2k < p < 2k + 2$, and write

$$s = \frac{p}{2} = k + \alpha, \quad 0 < \alpha < 1.$$

Then

$$\sum_{\substack{q \geq 1 \\ q \text{ odd}}} I_q(s) \geq I_1(s) - \sum_{\substack{q \geq 3 \\ q \text{ odd}}} |I_q(s)| > 0.$$

Since the prefactor in (15) is positive, $D(s) > 0$. By (3),

$$\|g_k\|_{L^p(\mathbb{T})} > \|f_k\|_{L^p(\mathbb{T})}. \quad \square$$

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A AI usage

Statement. The mathematical argument of this paper was produced by an artificial-intelligence system, not merely checked or edited by one. The system is the author’s personal AI research system; the language models behind it are Claude Opus 5 and GPT-5.6 Sol. The author has read and verified the argument of §§2–6, which is self-contained and can be checked by hand, and takes full responsibility for it, including for the parts a machine wrote. No formal verification is claimed. The rest of this appendix records what the system did, and where it was wrong.

The system runs about forty agent processes in parallel behind those two models, and assigns every adversarial re-check across model families. We record here what the system actually did, including where it was wrong, because a reader is entitled to know which parts of a proof were searched for by machine.

Where the contribution lies. We want to be plain about which step carries the weight. Most of the argument uses standard tools: restricting a Fourier series on $\mathbb{T}^2$ to a line (§2), the Mellin representation of J_0 and analytic continuation one mode at a time (§3), elementary bounds on Bessel functions for the tail (§5), and convexity of $\log \Gamma$ with a geometric comparison at the end (§6). The step we regard as the system’s real contribution is in §4, Propositions 4.1 and 4.2: recognising that the leading mode I_1 , whose two large Bessel orders are adjacent, collapses under Neumann’s product formula [1, Eq. 10.9.26] to a single Bessel product with the nonnegative weight $\cos \vartheta$; that the resulting integral $W(A)$ is a Weber–Schafheitlin integral [1, Eq. 10.22.56] with a closed form in ${}_2F_1$; and that after Euler’s transformation [1, Eq. 15.8.1] the hypergeometric series on the branch $1 < A < 2$ has nonnegative coefficients, so that positivity of the leading mode can be read off term by term. We do not claim that the combination of Neumann’s formula with the Weber–Schafheitlin integral is new; what the system did was to pick the right special-function identities out of the DLMF and see that they fit this problem.

What is not a contribution. Section 6 ends with comparisons of explicit rational numbers, such as the bound on $R(4, 0)$ in Lemma 6.2. These are finitely many multiplications of Gamma values at integers, written over the rationals so that a reader can check them by hand. The rational surrogates for π , $\sqrt{3}$ and e , and the target ratio $(3/4)(4/5)^{k-4}$, were chosen with the required inequality already in view; without that target one would simply try several choices. We record this as bookkeeping, not as an idea.

History. The first arXiv version of this paper followed a different route: an elementary reduction to the positivity of an explicit sum over an integer sequence σ_m , several parallel lines of attack on that inequality, and an exact-arithmetic machine verification on a finite range of k . The present version replaces all of that by the argument of §§2–6, which uses no computer. Two things survive from that exploration: the frame of the two-torus, onto which the successful line of attack was

rerouted from the circle, and the constant $9/16 = \min_{w>0}((1+w^3)/(1+w))^2$, which a lattice-path line produced independently and which reappears as the base of the factor $(9/16)^s$ in W_{low} .

The system’s output was wrong in four identifiable ways, and the corrections are part of the record. The first draft performed the analytic continuation at the level of the sum, where no summation-level identity exists — the Mellin identity converges absolutely only in a strip in which the Fourier coefficients are not summable over the lattice — so the continuation had to be redone one mode at a time, which is what §3.3 now does; this was a genuine gap and it is closed. A later re-derivation reported that the tail bound had dropped a factor, but the re-derivation was itself the error: the three Bessel orders sum to $2qN$, not $q(2N-1)$, with an explicit counterexample at $k=1$, $q=3$, $r=2.7724565$; the formula in the draft was correct and only the derivation around it was repaired. A standoff over the prefactor, in which two expressions appeared to differ by a factor $\Gamma(s+1)$, was traced not to an algebraic slip but to an object, σ_m , that the draft had never defined. A control function used the bound $|J_\nu(Ar)| \leq (2/(\pi Ar))^{1/2}$, which holds only for $\nu \leq 1/2$ and is exceeded by a factor 1.449 at $\nu=21$; it was replaced and the estimate reproved. The first draft of the earlier version also asserted that its argument needed no computer assistance; for that argument this was false, and the claim was deleted rather than weakened. The argument of the present version does not use a computer.

Parts of the argument were submitted to an interactive theorem prover (Lean 4 with Mathlib) in an exploratory attempt at formalisation; that attempt is incomplete, which is why no claim of formal verification is made above. The ancillary scripts distributed with this paper recompute the rational comparisons exactly and the analytic identities numerically; they are tools and are not part of the proof.